

The Art of Queuing: a Faster QuickPass System

Yao Xie⁽¹⁾, Chao Ma⁽²⁾, Hao Yu⁽³⁾, Jia Yao⁽⁴⁾

Dept of ⁽¹⁾ Electrical Engineering and Information Science ⁽²⁾ Statistics and Finance ⁽³⁾ Math. ⁽⁴⁾ Earth and Space Science
of University of Science and Technology of China

⁽¹⁾ xieyao@mail.ustc.edu.cn

Summary

This paper discusses performance of queuing systems, particularly QuickPass system, as well as design optimal operation strategy to minimize time cost function of amusement park tourists.

Firstly, we propose a reasonable evaluation criterion. Based on some practical regulations of FASTPASS in Disneyland, we model for two typical items in the park. The “phone booth model” was for single ride. We use queuing theory to construct a state transition equation, and then settle numerically that the optimal assigned return time to the second person should be 17.0min if mean serving time is 5min.

Our “scenic railway model” was detailed into: single QuickPass queue, two-queue model with both QuickPass Queue and ordinary StandBy Queue. For single QuickPass queue, we find the classical ‘Greedy Algorithm’ works well. Simulation shows it generally has 21.3% lower cost function than ordinary queuing. Also, we do large quantities of simulation to test performance of our model in different system capacity, cycle interval and other operation strategy, find optimization for system capacity as 60 seats, cycle interval as 4.67min, and compare efficiency FASTPASS with ordinary queues. We also design a scheme using trunking theory to statistically get queue length function, then choose return interval in the valley of the function. Simulation verifies it could generate more uniform tourist traffic. Finally we do solid theoretical verification for two – queue model.

Further, we discuss strength and weaknesses of our models, as well as proposal to improve future QuickPass systems. Conclusion is rendered in a non-technical paper.

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1 Introduction

Queuing - experience in the seemingly unending line, whether at phone booths, amusement parks, usually becomes trouble to people, which has been verified by some psychologists. It may also not only be concern of tourists, or customers, but also of the managements and the marketing offices. Popular rides attract large number of tourists forming long queues, which would not only be only annoying to customers, but also slow ordinary operation rate of the rides – the rate to gain profits! So develop a kind of “QuickPass” systems to reduce waiting lines become imperative.

What is QuickPass? Briefly, it is a kind machine, when insert your daily entrance ticket into the system, it will print an estimated time interval when you can come back to the amusement item, according to the data of the tourists come before you. Some can interactively help you choose the return time. So you can enjoy your time freely instead waiting in long queue, just come back in the indicated time on the slip. There are already real cases of QuickPass system in use, the most famous and efficient one is FASTPASS systems in Disneyland (Fig.1), which we would use as a good example to model later.



Fig. 1 FASTPASS
In Disneyland

However, how much of the problem can a QuickPass solve? How much advantage it has over ordinary lines? If so, what should be the criteria for evaluation, or more specifically, what should be the cost function that we seek to minimize? And how much trouble it can make? Nowadays some queuing system with improper algorithm would even cause more delay, their uneven arrangement cause complaints. How does this happen? Our mission in the paper is to settle those problems: reducing tourists' overall waiting time – assigned by QuickPass and in line queuing time, taking into consideration the two waiting time counts differently psychologically. In a word to make touring in park more enjoyable and time-saving. For this purpose, in this paper we constructed three models, with cautious simulations to test their viability.

Another important task is to find evaluating criteria, also a non-technical summary for amusement park executives to help them choose between alternatives queuing systems and “QuickPass”

2 Queuing Problem: Phone booth Model

Obviously our mission should get down to queuing problems, then knowledge about queuing theory become necessary. So we start our exploration with a simple queuing problem. In an amusement park, there is a kind of entertaining item, for example, the ride, it allows at most M tourists to play, in another word, when the number of tourists reaches to M , the tourist who comes late would not be allowed to play until at least one tourist come out. Here, we want to discuss the simplest situation: $M=1$, just like the phone booth.

2.1 Introduction to Queuing theory

Easily realized that the KEY POINT in this problem is to estimate how long the wait for a ride should REALLY be, so that to assigned a suitable returned-time for people.

As mentioned in any textbook on the theory of queuing, we use the following variables:

X: a random variance on the flow of people;

Y: a random variance on the service-time;

Z: the number of serve-station;

A: the maximal people served at one time;

Then, we use the symbol (X/Y/Z/A) represent a special kind of state of a queue. (here, X is assumed generally a Poisson flow, and Y obeys exponential distribution). There are two kind of queue-system in another criterion, that is the earlier a person come, the earlier he would be served (FCFS), the opposition of it is, obviously, the earlier a person come, the later he would be served (LCFS). Using classical queuing theory, we may get to know two important parameter – ALQ (Average Length of the Queue) and AWT (Average Wait-Time of a person).

2.2 Assumptions for phone booth problem

- In this model, X obeys the exponential distribution, $Z=1$, $A=1$. As what we have mentioned above, there is at most one person play on the ride at one time. Another, the QuickPass is a real time system, it could record some information about the person it has offered a slip, so that the QuickPass could make an optimal suggestion (offer a slip) to the next person according them. Well,
- Moreover, for the sake of evenness, in this model we have considered the principle that the earlier a person comes to the kiosk of the ride, the earlier he would be assigned to play on the ride, and so, it is FCFS.

In our model, the following variables are necessary:

Table 1 Variable Definitions

E_i	The time tourist(i) just be allowed to play on the ride (at this time, no one is playing on the ride).
P_i	the length of time tourist(i) play on the rides
W_i	the length of time tourist(i) waits in the line
T_w	the length of time of the ride work one day
t_i	the length of time assigned by QuickPass to tourist(i)

The direct purpose using QuickPass is to reduce people's time in the queue --- $w(i)$, at the same time, t_i should not be assigned too large, exceeding a certain limitation will cause people's unenjoyment. Considering these, we use the following goal-function:

$$\min \{U = c_1 t_i + c_2 W_i\} \quad \text{s.t.} \begin{cases} \sum_{j=1}^i T_j + t_j < T_w \\ t_i < T_{i+1} + t_{i+1} \end{cases}$$

For any $i=1, 2$, here c_1, c_2 is the attribution of a person, depending on his preference between T_i and W_i ,

so they are functions of i ; Further more, we define $A(i) = \sum_{j=1}^i T(j) + t(i)$.

2.3 System State Transition Equation

In the following we find solution to statistical equilibrium state of the queuing system, to discuss expectation of waiting time.

When tourist(n) comes into a park, the state of queue is as in Fig. 2:

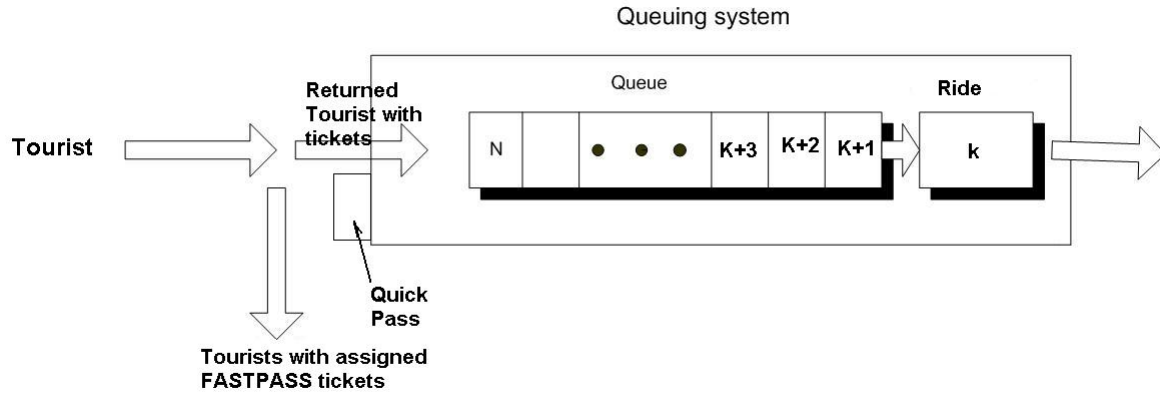


Fig. 2 The state of Queue of single ride

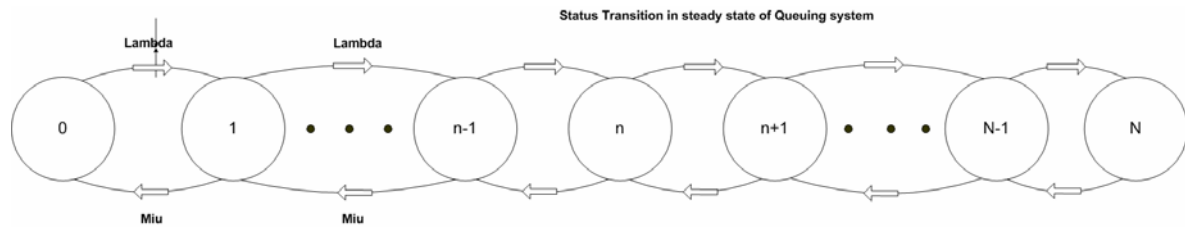


Fig. 3 State Transition of single QuickPass queuing system

Here, tourist(k) is playing on the ride. Or there is no tourist playing on the ride. So obviously in this kind of state, tourist(n) could wait

$$W_{n,k} = (E_k + \sum_{i=k}^n P_i - A_n)^* , \quad (x)^* = \begin{cases} x, & x > 0; \\ 0, & x \leq 0. \end{cases}$$

Thus, define Q_k is the probability of the occurrence of such queue-state above, then the goal-function is:

$$U = c_1 t_n + c_2 \sum_{k=1}^{n-1} Q_k W_{k,n}$$

and the sufficient and necessary condition the state occurs is that :

$$E_k + P_k > A_n > E_{k-1} + P_{k-1}$$

We define

$$\begin{aligned}
 Q_{k,1} &= \text{probability of } E_k + P_k > A_n, \\
 Q_{k,2} &= \text{probability Of } E_{k-1} + P_k + P_{k-1} > A_n, \\
 &\dots \dots \\
 Q_{k,s} &= \text{probability Of } E_{k-s+1} + P_k + \dots + P_{k-s+1} > A_n.
 \end{aligned}$$

More, we can realize that

$$E_k = \begin{cases} A_k, A_k > E_{k-1} + P_{k-1}, \\ E_{k-1} + P_{k-1}, A_k \leq E_{k-1} + P_{k-1} \end{cases}$$

Thus $Q_{k,i}$ satisfies

$$\begin{aligned}
 Q_{k,i} &= (1 - Q_{k-1,i}) + Q_{k-1,i} Q_{k-1,i+1}, \quad i \leq k \\
 Q_{k,k} &= \text{probability of } \left(\sum_{i=1}^k P_i > A_n \right) \quad (E_1 = 0)
 \end{aligned}$$

Here we know P_i is submit to the exp-distribution with parameter λ and they are independent, so

$Q_{k,k}$ could be calculated (a function of $t_1, t_2, \dots, t_k$).

2.4 Optimization of Returned time assigned by QuickPass

Now, we can solve the goal-function and t_i . For example, when person(1) comes into the entrance of the ride-gate, he could play on the ride directly, so the QuickPass need not offer a slip to he ($t_1 = 0$). After T_2 (minutes etc), person(2) comes, because $t_1 = 0$, the goal-function is

$$\begin{aligned}
 U_{t_2} &= c_1 t_2 + c_2 (P_1 - T_2 - t_2)(1 - N(T_2 + t_2)) \\
 T_2 + t_2 &< T_w; \quad (N(x) = \int_{t=0}^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}, x > 0)
 \end{aligned}$$

Fig. 3 is the image of U_{t_2} in Matlab 6.5. We can see that there is really a minimal-point of U_{t_2} , we could solve the approximate minimal value of U_{t_2} and the correspond t_2 .

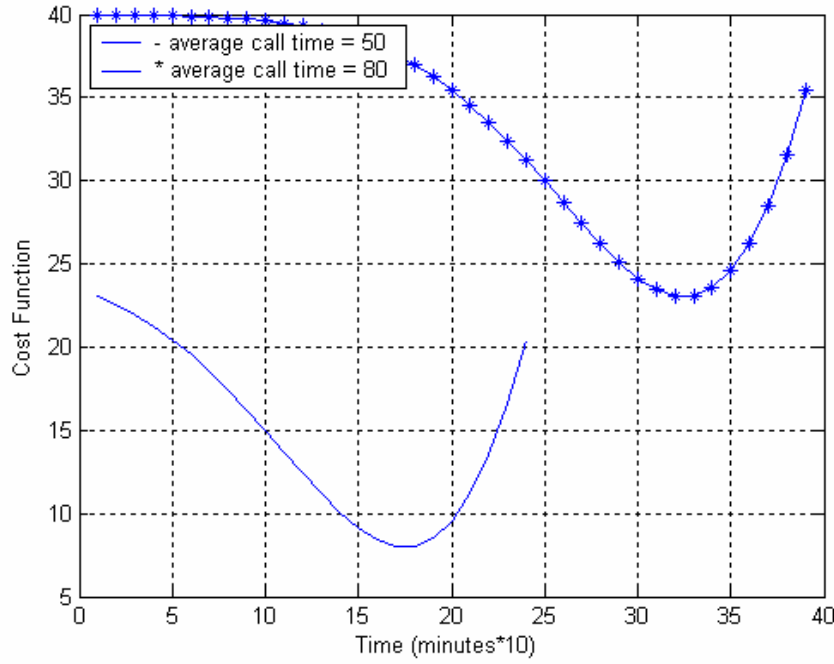
Fig. 4 Δt assigned by QuickPass vs. Cost Function $c_1 = 2, c_2 = 1$ 

Table. 2 the minimal value of U

Average call time(min*10 ³)	Min U	t_2
50	8.0516	17.0
80	23.0516	32.5

Further, assume x, y are independent random variables, which obey exponential distribution, according to the theory of probability, the random variable $z = x + y$ obeys exponential distribution with parameter λ^2 . Let $G(x)$ is the distribution function of z , in the same way, we could get the goal-function about person(3).

$$\begin{aligned}
 W_3 &= (1 - N(T_2 + T_3 + t_3))(P_1 - T_2 - T_3 - t_3) \\
 &+ N(T_2 + T_3 + t_3)(N(T_2 + t_2)(1 - N(T_3 + t_3 - t_2))(t_2 + P_2 - T_3 - t_3) \\
 &+ (1 - N(T_2 + t_2))(1 - G(T_2 + T_3 + t_3))(P_1 + P_2 - T_2 - T_3 - t_3)) \\
 &(1 - N(T_2 + t_2))(1 - G(T_2 + T_3 + t_3))(P_1 + P_2 - T_2 - T_3 - t_3)
 \end{aligned}$$

The goal-function is:

$$\min \{U = c_1 t_3 + c_2 w_3\} \quad \text{s.t.} \quad \begin{cases} T_2 + T_3 + t_3 < T_w \\ t_2 < T_3 + t_3. \end{cases}$$

Likewise, we can draw its image and solve the approximate minimal value of U in Matlab. There is no difficult that we could get the expression of $w_4, w_5, \dots$, and get the minimal value of U .

Generally speaking, the capacity of the item is often constant, and >1 , the arriving time of the tourist is subject to certain distribution, the serviced time of a tourist is also fixed.

3 Scenic Railway Model

How you ever ride a Scenic railway? – and even been to Disneyland to use its FASTPASS? If so you would understand scenic railway subject to another kind of queuing problem.

3.1 QuickPass or Standby?

Classification

Here, we divide the way by which the customers wanting to visit the attraction, which allows to use the “QuickPass” system, into two kinds, one is the usual line called “StandBy” line, and another is “QuickPass” line, just like Disney’s FASTPASS in Fig. 5. Of course, each customer has the right to choose which line to enter the attraction, but from the whole to see, the customers will be divided three parts, the following is:

- Because of the limited amount of the “QuickPass” tickets, it is possible that the customers can not buy the “QuickPass” ticket; they only can to join in the “StandBy” line if they want to continue to play the attraction. Assumbaly the case does not

often occur, it occurs in high day like holiday, weekend. And the case can be expected, so the park easily deals with some problems caused because of the large amount of peoples being in “standby” line, they do not buy QuickPass ticket, for example, the executives can increase the amount of the “QuickPass” ticket, they can also add the opening Time of these popular attractions. In fact, people Relatively easily submits to the long waiting time during holiday or weekend, because they have adequate preparation before they go out of home.

- The world is so large that it is possible that anything can happen, so there are some people who do not believe the new method-----“QuickPass” system though it can really do better, thus the only way to the entrance is to wait in line. We do not inference their favor, but we must service their that favor, and we can either not make them wait too long time though they do not trust our new technology, so it is necessary to assign the usual line.
- The remain is that the “QuickPass” ticket -holding customers, they receive the slip and be told the

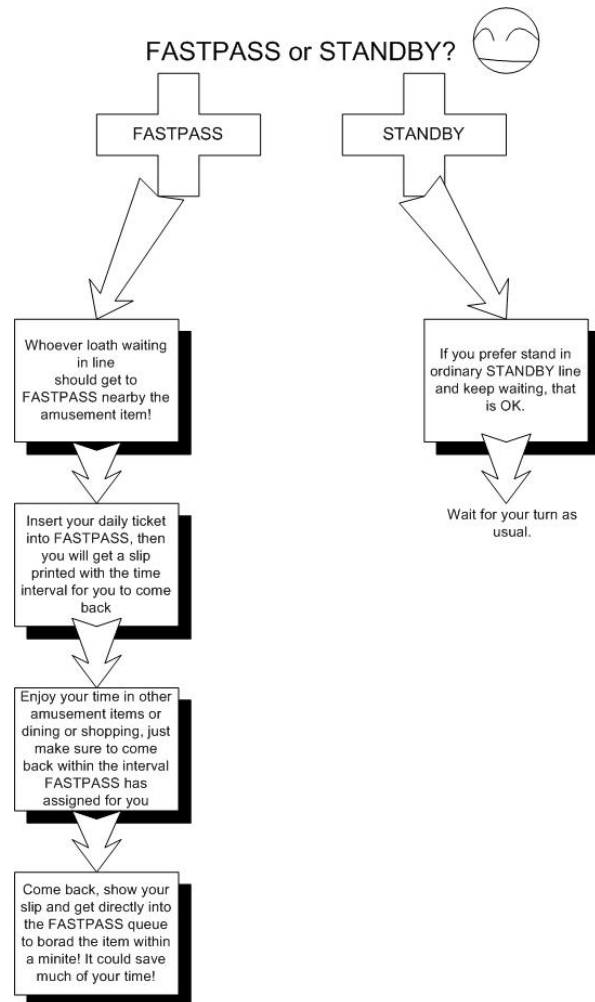


Fig. 5 Two-queue system: FASTPASS

return time ,so they can free visit the other part of the park during the interval, they can also use the interval to play the other item, do any other things.

Then we could assume the usual line and the “QuickPass” line must open at the same time, but they should not open dependently with each other, thus the problem is how to assign the two lines, that is, how do we distribute the customers into the two lines? And, in the last case in the discussion above, the following case can emerge, the customers can see the fewer amounts of people in the usual line, so they can give up the “QuickPass” queue, and go to join in the “standby”.

In our model, we take the scenic railway as an example. First, to make the problem simplify, we only adopt one queue, then, we design the following model.

3.2 One-queue Model

3.2.1 Assumptions

- The amusement park opens from 8:00 to 18:00. At a certain time t during this work time, the number of the tourists arriving at an amusement machine, such as scenic railway, has a Poisson distribution with Parameter $\lambda_{(t)}$.

$$P(k) = \frac{(\lambda_{(t)}t)^k e^{-\lambda t}}{k!}$$

- There is only one Queue at the scenic railway. When a tourist arrives, facing a long queue, QuickPass may suggest him go around and return at another time, when the queue may be shorter. He stands at the end of the queue, probably not waiting too long.
- The cycle interval of the scenic railway is a constant T . The points during a day' worktime at which the railway starts and stops are fixed. The railway must start off at the start time despite whether it is full.

Our goal is to determine the time suggested by the QuickPass system for the tourist's coming back. It must satisfy 1)the queue may be relatively short, and 2)this return time should be as close to the tourist's arrival time as possible.

3.2.2 Variable Definitions

Table. 3

Variable	Definition
Δt	The work time during a day is divided into m periods, each equal to Δt
$\hat{t}_i$	The beginning of period i . i is from 1 to m .
l_{ij}	The length of the queue at the scenic railway at the beginning of period i in day j .
Δt_1	The difference between the time suggested by the QuickPass for next arrival and this moment ,during which the tourist could go around for some other amusement

Δt_2	The length of the time a tourist waiting in line.
M	The system capacity .In a Scenic railway it is the number of the seat.
c_1	The cost the tourist pays in a unit time when going around in assigned time interval.
c_2	The cost the tourist pays in a unit time when waiting in line.
$\bar{l}$	The mean of the length of the queue at a certain point in a day
t^i	The moment that the ith passenger arrived this day.
$l(t)$	The length of the actual queue at time t this day. We assume $l(0) = 0$.
$\bar{l}(t)$	Predictive length of queue at time t this day.

3.2.3 Sample Space to Determine PDF Parameter

Using the information of the tourists during the past 100 days, this could be easily gathered from the tourist's tickets (may be a IC card, which would keep the tourist's information in a central database system of the park). Then we can get the number of l_{ij} , where i is from 1 to m and j is from 1 to 100. They form a matrix of Sample Space.

$$\begin{pmatrix} l_{11} & l_{12} & \dots & \dots & l_{1,100} \\ l_{21} & l_{22} & \dots & \dots & l_{2,100} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ l_{m,1} & l_{m,2} & \dots & \dots & l_{m,100} \end{pmatrix}$$

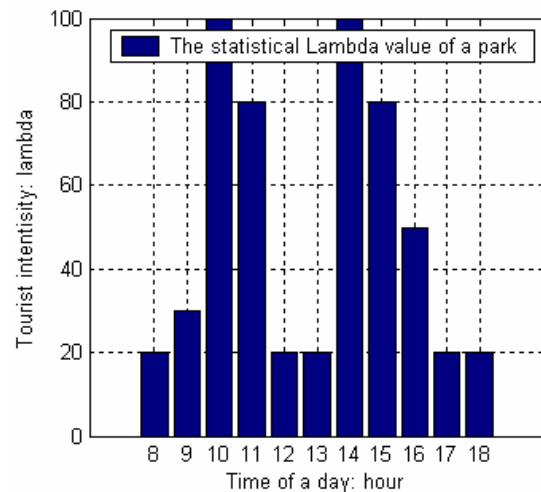


Fig. 6 Statistical curve of $\lambda(t)$

Because at a certain time t during this worktime,

the length of the queue at a amusement machine, such as scenic railway, has a Poisson distribution

With Parameter $\lambda_{(t)}$, we can use $l_{i,1}, l_{i,2}, \dots, l_{i,100}$ to estimate the parameter $\lambda(t_i)$, as shown in Fig. 6 where

i is from 1 to m . Then if the interval approaches to infinite small, then $\lambda(t)$ become continuous function, the arrival events process obeys Nonhomogenous Poisson Process.

3.2.4 Average Real-time Queue Length – Erlang Formula C

So how to calculate $\bar{l}_{(t)}^{1,1}$ in statistical mean. Since tourists are assumed to arrive with a Poisson distribution, and it is further assumed that there could be infinite number of tourists waiting in line, then

Erlang C formula [Stat90] could be used to describe the grade of service (GoS) as the probability that an arbitrary user will be blocked and wait in the queue. The Erlang C formula is given by [2], as follows.

$$\Pr[blocked] = \frac{A^C}{A^C + C!(1 - \frac{A}{C}) \sum_{k=0}^{C-1} \frac{A^k}{k!}} = GoS(Grade_of_service)$$

Then tourists arrival rate should be traffic intensity, which is measured in Erlang in classical Queuing theory. One Erlang represent the amount of traffic intensity carried by a system whose capacity is completely occupied. , where A is the current traffic intensity, which is just the $\lambda(t)$ in our model.

C is system capacity, and the possibility that the delayed call is forced to wait for more than t seconds in the queue is given by

$$P[wait > t | blocked] = e^{-\frac{(C-A)t}{H}}$$

Then we can get

$$\overline{l_{(t)}} = \Pr[blocked] \times A \times \Delta t$$

Actually, we could use Erlang C Graph (Fig. 7) to calculate $\Pr[blocked]$.

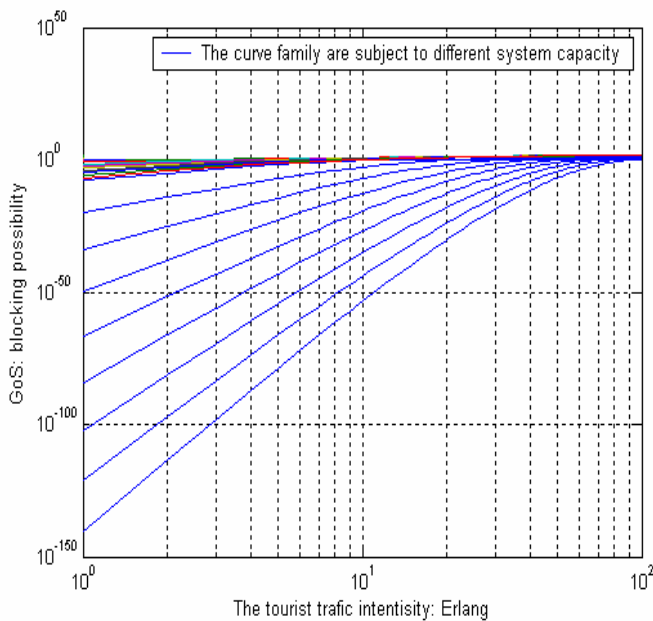


Fig. 7 Graph of Erlang C PDF

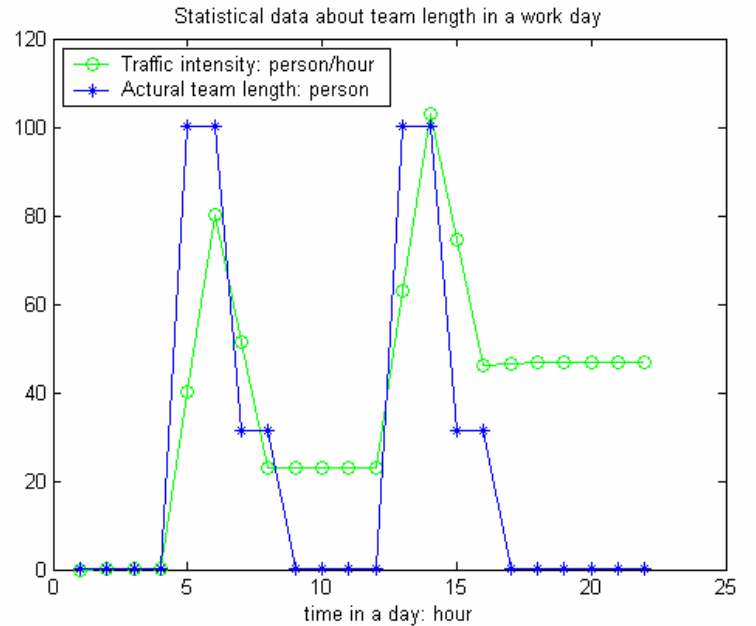


Fig. 8 Statistical curve of Real-time queue length

And the result is shown in Fig. 8. The graph could be used to determine optimal returning time interval in the following model.

3.2.5 Cost Function Minimization Algorithm

To obtain return time interval Δt_1 , which is proposed by QuickPass to the tourist, we should define the

cost function Z like this:

$$Z = c_1 \Delta t_1 + c_2 \Delta t_2$$

And now, our objective is to find the value of Δt_1 to minimize Z .

Because the longer a tourist waits, the more fidgeted he becomes, c_1 or c_2 increases while Δt_1 or Δt_2 . When Δt_1 or Δt_2 is beyond a certain number, for example, $\Delta \tilde{t}_1$ or $\Delta \tilde{t}_2$, the passenger will be so fidgeted that he can not put up with any longer (The result from the research states that the customers become boring when they wait time beyond 10 minutes, and after 20 minutes, they behavior more evident, and if the waiting time is more 40 minutes, they often leave with the face red). The cost the tourist pays in a unit time while waiting will increase as the total time he have waited increases. That has something to do with the Law of Increasing Marginal Utility in Micro-Economics. Then we define c_1 and c_2 like this:

$$c_1 = \begin{cases} \alpha_1 + \alpha_2 \Delta t_1 & (\text{when } \Delta t_1 < \Delta \tilde{t}_1) \\ +\infty & (\text{when } \Delta t_1 \geq \Delta \tilde{t}_1) \end{cases}$$

$$c_2 = \begin{cases} \beta_1 + \beta_2 \Delta t_2 & (\text{when } \Delta t_2 < \Delta \tilde{t}_2) \\ +\infty & (\text{when } \Delta t_2 \geq \Delta \tilde{t}_2) \end{cases}$$

When the scenic railway is running, the first M tourists in line must wait until it stops. The mean of the time they wait is $\frac{T}{2}$. So

$$\Delta t_2 = \left\lfloor \frac{\bar{l}}{M} \right\rfloor T + \frac{T}{2}$$

Let $\bar{l} = \bar{l}_{(t^i + \Delta t_1^i)}^{1,i}$. Here $\bar{l}$ is our prediction of the queue length at the time suggested by the QuickPass system.

So we get:

$$Z = c_1 \Delta t_1^i + c_2 \left(\left\lfloor \frac{\bar{l}_{(t^i + \Delta t_1^i)}^{1,i}}{M} \right\rfloor T + \frac{T}{2} \right)$$

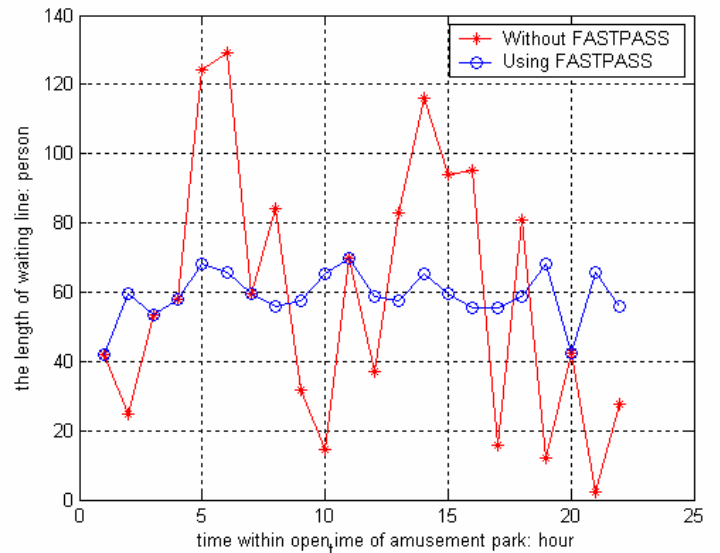


Fig. 9 The length of waiting line in two conditions

We can find the value of Δt_1^i that minimizes Z .

Considering the actual queue now, if the queue at the time suggested by QuickPass is much probably too long that the cost of using the QuickPass ticket is even larger than the cost of waiting in line from now on.. Our system would let the passenger stand in line from now on, rather than get FASTPASS ticket to go around. So we let

$$\widehat{Z} = \min \left\{ Z, c_2 \left(\left\lfloor \frac{l(t)}{M} \right\rfloor + \frac{T}{2} \right) \left(\left\lfloor \frac{l(t)}{M} \right\rfloor T + \frac{T}{2} \right) \right\}$$

$$\bar{I}^i = \begin{cases} \bar{I}_{(t^i + \Delta t_i)}^{1,i} & (\text{if } \widehat{Z} = Z) \\ l(t^i) & (\text{if } \widehat{Z} = c_2 \left(\left\lfloor \frac{l(t)}{M} \right\rfloor + \frac{T}{2} \right) \left(\left\lfloor \frac{l(t)}{M} \right\rfloor T + \frac{T}{2} \right)) \end{cases}$$

If $\bar{I}^i = \bar{I}_{(t^i + \Delta t_i)}^{1,i}$, that means QuickPass suggests the tourist go around and return later, so curve $\bar{I}_{(t^i + \Delta t_i)}^{1,i}$ should change into curve $\bar{I}_{(t^i + \Delta t_i)}^{1,i+1}$, which is faced by the $i+1$ th tourist today at his first arrival. But the former tourist may not return exactly at the time suggested by QuickPass. We assume that it obeys Normal Distribution- $N(t^i + \Delta t^i, \delta^2)$. And the probability density function is $f^i(t)$. So

$$\bar{I}_{(t^i + \Delta t_i)}^{1,i+1} = \bar{I}_{(t^i + \Delta t_i)}^{1,i} + f^i(t)$$

Else if $\bar{I}^i = l(t^i)$, it means QuickPass suggests the person wait in line from now on. We assume that the tourist actually stands in line with the probability 1. So there is no need to make any change on curve $\bar{I}_{(t^i + \Delta t_i)}^{1,i}$. That is to say:

$$\bar{I}_{(t^i + \Delta t_i)}^{1,i+1} = \bar{I}_{(t^i + \Delta t_i)}^{1,i}$$

At the beginning of everyday worktime, we use curve $\bar{I}_{(t)}^{1,1}$.

3.2.6 Time Sequence Test to determine Sample renewal

Our curve $\bar{I}_{(t)}^{1,1}$ is based on the data in the past 100 days. But because of the effect of Quickpass's suggestion, the habit of tourist and the traffic flow during a day will change. The former curve may not predict exactly any longer. So we must build a new curve. We should consider when to add new data to build a new curve and how to build it.

When we use curve $\bar{I}_{(t)}^{1,1}$ for 10 days, that is to say, today is Day 111. We use the matrix

$$\begin{bmatrix} l_{11} & l_{12} & \dots & \dots & l_{1,100} \\ l_{21} & l_{22} & \dots & \dots & l_{2,100} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ l_{m,1} & l_{m,2} & \dots & \dots & l_{m,100} \end{bmatrix}$$

to get an arrow A1,

$$A1 = \begin{bmatrix} \tilde{l}_1 \\ \tilde{l}_2 \\ \dots \\ \dots \\ \tilde{l}_m \end{bmatrix}$$

where $\tilde{l}_k = \sum_{j=1}^{100} l_{ij}$. $\tilde{l}_k$ is the mean of the length at the beginning of a certain period (which is $\hat{t}_k$)

during a day from Day 1 to Day 100. Then we use the data in the latest 10 days, from Day 101 to Day 110, which form a matrix of Sample Space.

$$\begin{bmatrix} l_{1,101} & l_{1,102} & \dots & \dots & l_{1,110} \\ l_{2,101} & l_{2,102} & \dots & \dots & l_{2,110} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ l_{m,101} & l_{m,102} & \dots & \dots & l_{m,110} \end{bmatrix}$$

to get an arrow B1.

$$B1 = \begin{bmatrix} \tilde{l}'_1 \\ \tilde{l}'_2 \\ \dots \\ \dots \\ \tilde{l}'_m \end{bmatrix}$$

Here $\tilde{l}'_k = \sum_{j=101}^{110} l_{ij}$. $\tilde{l}'_k$ is the mean of the length at the beginning of a certain period (which is $\hat{t}_k$)

during a day from Day 101 to Day 110. Then we can get the distance between Arrow A1 and Arrow B1.

$$d = \sqrt{\sum_{k=1}^m (\tilde{l}'_k - \tilde{l}_k)^2}$$

If $d \leq d_0$ (d_0 is a certain number), it means that the habit of tourists and the traffic flow during a day have not changed obviously. So there is no need to change curve $\bar{l}_{(t)}^{1,1}$. Else if $d > d_0$, it means that the habit of tourists and the traffic flow during a day have changed obviously. So we should change curve $\bar{l}_{(t)}^{1,1}$ into curve $\bar{l}_{(t)}^{2,1}$.

But how to build curve $\bar{l}_{(t)}^{2,1}$. We use the data from Day 101 to Day 110 to build curve $\bar{l}_{(t)}^{2,1}$ just the way as

we use the data from Day 1 to Day100 to build curve $\bar{l}_{(t)}^{1,1}$. Likewise we change curve $\bar{l}_{(t)}^{2,1}$ into $\bar{l}_{(t)}^{3,1}, \bar{l}_{(t)}^{3,1}$ into $\bar{l}_{(t)}^{4,1}$, and so on.

3.2.7 Numerical Solutions to the Model

This model uses the statistic data to predict the transformation of the queue length during a day's work time. It suggests the tourists at a rush time come back at a low time. So "the traffic flow" may become smoother. Our test indicates that: The value interval of the actual length during a day's worktime is 126.7744 without the QuickPass system. And it is 27.7648 when using the QuickPass system. The percentage of the diminished value interval is 78.10%. The standard variance of the length before using the QuickPass system is 37.7856. And it is 7.2733 when using the QuickPass system. The decreased standard variance using the QuickPass system is 80.75%

Tourists coming back have no priority to tourists who arrive first. It seems to be not just. Although the one-queue model can roughly design the "QuickPass" system, the fact that the standby line exists is not ignored, so the two-queue model is more suitable to the real condition.

3.3 Two-queue Model

Obviously, here, we will consider 2 lines: "standby" line and "QuickPass" line. But, there is a problem that must be spoken clearly, that is, the usage of creating the "standby" line, its necessity has been stated above.

3.3.1 Two-queue Model not-considering cost of STANDBY Queue

The model is based on that we do not consider the benefit of the customers from the "standby" line. So, the following case is out of the range considered: some customers may join in the "standby" line after s/he gets the slip, later maybe s/he entrances the item by the "standby" line. Here, we assume the customers considered in our issue entrance the attraction all by the "QuickPass" line.

3.3.1.1 Assumptions:

- The customers' coming time correspondents Passion distribution, and any 2 customers' coming times are dependent each other.
- The customer's service is subordinate to the "First come First paly", that is, the return time of the customer first going to the kiosk near the attraction to insert the daily park entrance ticket will be assigned earlier.
- How many people will be told to return during the same interval is determined only by the capacity of the item, we assign 80% of the capacity of the item in the same return interval, but ,to prevent the case as the system first tells the customer return after 4 hours, but later the system tells the people return on the same item only after 1 hour, we should shorten the return time interval to make the customers distribute uniformly among all intervals.
- "QuickPass" system is to some extent dividing the usual line into some short queues whose length is prescribed. We divide all the opening time into the same length of interval, here, we assume the whole opening time of the item is 8 hours, and divide 8 hours into 8 intervals, the length of each interval is 1 hours. Thus, the system offers 8 suppositional lines, the customers only wait in their own suppositional line.

- The park issues the amount of “QuickPass” tickets is only a part of the max capacity of the item, here, we assume the part is 80% according to the rule of Disney park’s FASTPASS. That is, the system only assigns 80% of the item’s capacity at most in the same interval.
- The customer’s assignment is subject to the **Greedy Algorithm** : first filling in the prior interval until the number of the prior one’s people-assigned are more than the fixed amount definite above, the following is filling in the coming latter one.
- If the customer returns after the return time-told, s/he can insert her/his ticket and get a new return time slip or go to the usual line, s/he will not receive any priority.

3.3.1.2 Tourists return time assignment – Greedy Algorithm

As these problems above, here, we adopt the rule: each slip will tell the customer a span of the return time, the customer can return at any time during the interval, which has been defined above, but the does not state that the customer can paly the item once s/he returns, because the item has itself own operating rule, for example, a riding item may open once 30 minutes. If the amount of the customers being in the “QuickPass” line is less than the capacity of the item once when the riding item operates at once, thus we can use the customers from the “standby” line to make up for the margin. So, we do not open 2 sets each item at once, one belongs to the usual line, another belongs to the “QuickPass” line, instead of, we only open 1 set each item, and we definite a time or several times during each returning span, which is often several minutes before the item begins, and we prescribe the orderliness that if the customers returning before the definitive time, thus they will be assigned to play the coming bout, and the margin will be filled with the customers being in the usual line between the definitive time and the item opening time.

The idea above can describe the following with mathematic language.

We assume that k persons $\{h_1, h_2, \dots, h_k\}$ are assigned to return during the same returning

interval $(t_1, t_2]$ by the “QuickPass” system, and

we can choose the length of the interval

$(t_1, t_2] : |t_1 - t_2|$ to make the item opens 2 times

during the interval, thus we can only define a time

t_3 as the definitive time above.

We assume that the time a customer returning during the interval is subject to the Beta distribution $Be(\alpha, \beta)$.

For a customer, the graph of the possibility of s/he returns during the interval is: fig. 9.

Based on the assumptions, we now divide the opening time into 8 parts as following:

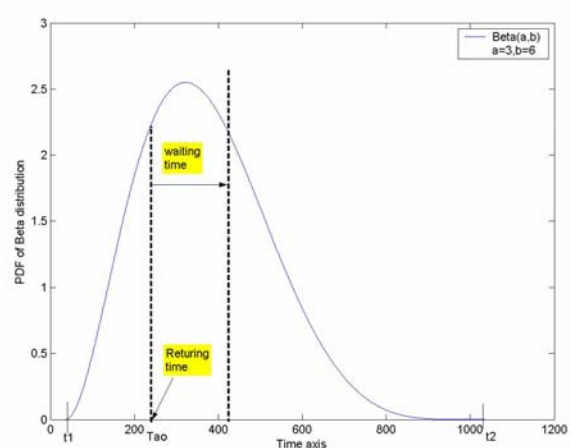


Fig. 9 the tourist return time distribution within the interval

$$(b_0, b_1], (b_1, b_2], \dots, (b_7, b_8], \text{ here, } b_i - b_{i-1} = 1, \text{ and, } i = 1, 2, \dots, 8$$

We divide all needing time to entrance the item for a customer into 2 parts, one is the interval between the moment s/he inserts the ticket and the time s/he returns, another is the waiting time which s/he returns and waits to entrance the item. For the k person, the prior one is stated with t_{k_1} , and the latter one is t_{k_2} .

3.3.1.3 Obtaining Expectation of Cost Function

Well, we assume the k person returning time is τ_k , and her/his returning time $\in (b_{j-1}, b_j]$, for receiving the simpler operation, we assume that the item only begins once during each interval, thus, t_3 above is the largest point during the interval. Thus, according to the graph above, we have $t_{k_2} = b_j - \tau_k$, here, $\tau_k < b_j$.

And $t_{k_1} = \tau_k - t_k$, here, t_k - the time that the k person inserts the ticket, the value of t_k is told by the

customer inserts the ticket into the system. Now our objective is: $E \left\{ \sum_k (c_{k_1} t_{k_1} + c_{k_2} t_{k_2}) \right\}$, here, c_{k_1} is the

important degree because of the cost of the customer can spend freely time. Because some customers want to play the item at once, the system tells that they can have time to do any other things, but they do not think it is a good thing, thus t_{k_1} also causes the cost. c_{k_2} is the important degree because of the waiting

cost in the line, it is necessary to each customer. Obviously, generally speaking, $c_{k_1} < c_{k_2}$, and

$c_{k_1} + c_{k_2} = 1$, c_{k_1}, c_{k_2} is constant.

$$\begin{aligned} E \left\{ \sum_k (c_{k_1} t_{k_1} + c_{k_2} t_{k_2}) \right\} &= \sum_k \left\{ E(c_{k_1} t_{k_1} + c_{k_2} t_{k_2}) \right\} \\ &= \sum_k \left\{ c_{k_1} (E\tau_k - Et_k) + c_{k_2} (b_j - E\tau_k) \right\} \\ &= \sum_k \left\{ (c_{k_1} - c_{k_2}) E\tau_k - c_{k_1} t_k + c_{k_2} b_j \right\} \\ &= \sum_k \left\{ (c_{k_1} - c_{k_2}) c - c_{k_1} t_k \right\} + \sum_k c_{k_2} b_j \quad (*) \end{aligned}$$

here, $E\tau_k = \frac{\alpha}{\alpha + \beta} = c = \text{constant}$, because τ_k is subject to Beta distribution, $Be(\alpha, \beta)$, b_j is relevant

to k . Based on the assumptions, we know if the amount of the customers is sufficient, thus there are $0.8M$ persons are assigned in each returning interval, here, M is the max capacity of the item. If it is not sufficient, there are $0.8M$ persons in the prior some intervals, and the following interval is less than $0.8M$.

Now, for simplifying $(*)$, we can assume c_{k_1}, c_{k_2} is not relevant to k , that is, $c_{k_1} \rightarrow c_1$,

$c_{k_2} \rightarrow c_2$, thus, $(*)$:

$$D + c_2 \sum_k b_j = D + 0.8M \sum_{j=1}^n b_j + \left(\sum_k -0.8Mn \right) c_2$$

here, $D = \sum_k \{(c_1 - c_2)c - c_1 t_k\} = \text{constant}$. n - the first n intervals all have $0.8M$ persons, but the $n+1$ interval is less than $0.8M$ persons.

In modeling above, we can only consider the benefit of the customers from the “QuickPass” line, but, from the angle of the whole profit of the park to see, the park must consider the good of the customers from the “standby” line, so the following is based on it.

3.3.2 Two-queue Model Considering Cost of STANDBY Queue

Because of the rule that the customers holding the “QuickPass” ticket have the priority of the service to the customers being in the usual line when they are waiting at the same time, the tourists in the QuickPass line are assigned in advance. The suggestion made by the QuickPass system is a period, instead of a point during a day’s worktime. If the tourist comes back beyond this period, the QuickPass ticket will be invalid.

If a tourist gets a QuickPass ticket, he may benefits from it. But some waiters in the standby line will wait longer.

Our goal is to determine the period suggested by the QuickPass system, which makes the tourist benefits from the QuickPass ticket much, and does not make too many waiters in the standby line wait longer.

3.3.2.1 Variable Definition

Table. 4

Variable	Definition
$\bar{l}s(n\Delta t)$	The mean of the standby queue length at the beginning of the n th period during a day’s worktime
$\bar{l}q(n\Delta t)$	The mean of the QuickPass queue length at the beginning of the n th period during a day’s worktime
$\tilde{l}s(t)$	The actual length of the standby queue at the time t .

3.3.2.2 Tourist Return Time Algorithm Optimization – marginal effects analysis

We can use the data in the past 100 days to build curve $\bar{l}s(n\Delta t)$ and curve $\bar{l}q(n\Delta t)$. Consider a tourist arrives at the time t , in the n th period. At this moment the actual length of the standby queue is $\tilde{l}s(t)$. If he

waits in the standby line from now on ,he will wait $(\frac{\tilde{I}s(t)}{W})T + T_1$, where $W = \frac{\sum_{n=1}^m (M - l_{q(n\Delta t)})}{m}$,and

T_1 is the mean of the difference between the moments of his arrival a next start time of the railway. Here we divide $\tilde{I}s(t)$ with W , instead of M ,because every cycle interval of the railway, the number of the tourists in the standby line who could get on the railway can not be up to M if the QuickPass queue is not empty.

If the tourist gets a QuickPass ticket, it will suggest he come back in the $n + \Delta n$ th period. When he comes back, he may wait T_2 in the QuickPass line. Here T_2 is the mean of the difference between the moment he comes back and the next start time of the railway. So if he gets a QuickPass ticket, he may reduce the time waiting in line by $(\frac{\tilde{I}s(t)}{W})T + T_1 - T_2$, but he must go around for nearly $\Delta n \Delta t$.So his cost will be reduced by

$$\Delta Z = c_1 \Delta n \Delta t + c_2 (T_2 - (\frac{\tilde{I}s(t)}{W})T - T_1)$$

Here we assume that c_1 and c_2 are constant for simple.

If the tourist arrives in the n th period, QuickPass have $m-n$ choices when it determines in which period he can come back. So let Δn be from 1 to $m-n$, then we can get $\Delta Z_1, \Delta Z_2 \dots \Delta Z_{m-n}$.Maybe some of them are even negative , which means taking a QuickPass ticket is not better than waiting in the standby line for the person.

Consider there is a positive ΔZ .It means the person could benefit from the QuickPass. But if he gets the QuickPass ticket, the cost of some other waiters in standby line will increase.

The $(M - \bar{l}q(n\Delta t))$ th tourist, the

$(2M - \bar{l}q(n\Delta t))$ th tourist, and so on, in the

standby line will wait T longer .The number of these unlucky waiters is equal to

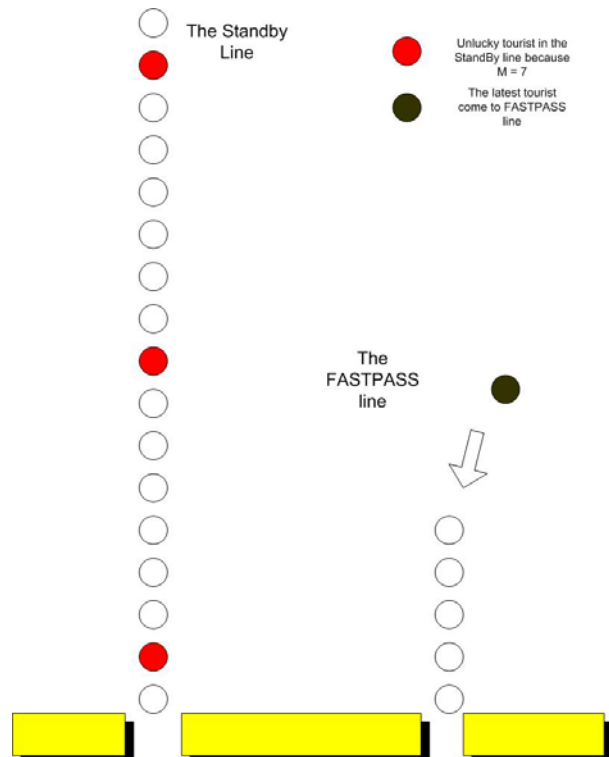


Fig. 10 Demo of Two-Queue system operation

$$\left\lfloor \frac{\bar{I}S_{((n+\Delta n)\Delta t)} - (M - \bar{I}q_{((n+\Delta n)\Delta t)})}{W} \right\rfloor. \text{ So the total cost of other waiters in standby line will be increased}$$

by :

$$\Delta Z' = c_2 T \left\lfloor \frac{\bar{I}S_{((n+\Delta n)\Delta t)} - (M - \bar{I}q_{((n+\Delta n)\Delta t)})}{W} \right\rfloor$$

Also we let Δn be equal to from 1 to $m-n$ respectively, then we can get $\Delta Z'_1, \Delta Z'_2 \dots \Delta Z'_{m-n}$.

Let, $Y = \min \{ \Delta Z'_k - \Delta Z_k \}$,

where k is from 1 to $m-n$. Then we can find the value of Δn that generates Y . This value is just what QuickPass should suggest to the person in order to increase people's enjoyment of the amusement park most.

The number of the valid QuickPass ticket indicating a certain period could not become beyond a constant $\tilde{M} (\tilde{M} < M)$. It insures that the length of the QuickPass queue is always smaller than the system capacity M .

At a certain point of a day's worktime, if there have been already i person who got the quickpass's tickets, curve $\bar{I}q^{1,i}(n\Delta t)$ should be changed into $\bar{I}q^{1,i+1}(n\Delta t)$.

If the habit of people and the traffic flow during a day's worktime have changed obviously, we should change curve $\bar{I}q^{1,1}(n\Delta t)$ into $\bar{I}q^{2,1}(n\Delta t)$, $\bar{I}q^{2,1}(n\Delta t)$ into $\bar{I}q^{3,1}(n\Delta t)$, and so on.

This model considers the cost of all the tourists in the park, including ones in the standby line and in the QuickPass line. So it can make all the tourists enjoy the amusement park better, not only the tourists in the QuickPass line.

4 Simulation and Model Analysis

4.1 Generation of Sample Space

We use $\lambda(t)$ to generate the sample space used in simulation, which is a Nonhomogenous Poisson Process, i.e. a set of 2000 random points indicates the arrival time of tourists, which was shown in fig. 10. However, we use another simpler and better equivalent method to generate data. According to Theorem 2.2 [Stochastic Process, USTC, 1991], samples which is randomly chose from uniform interval $U[0,1]$ could to use to simulate Poisson Process, such as shown in Fig.11.

4.2 Greedy Algorithm Test

Since the situation of queuing of FASTPASS too complicated to have a simple description, find certain parameters to optimize an explicit cost function become a N.S (Non-solution) problem. Fortunately, we could use computer simulation to approach a quasi-optimal algorithm. Not to lose generality, we use a simple ‘Greedy algorithm’ as described before in our simulation, and Fig 12 shows the scenic railway shift arrangement for arrived tourists.

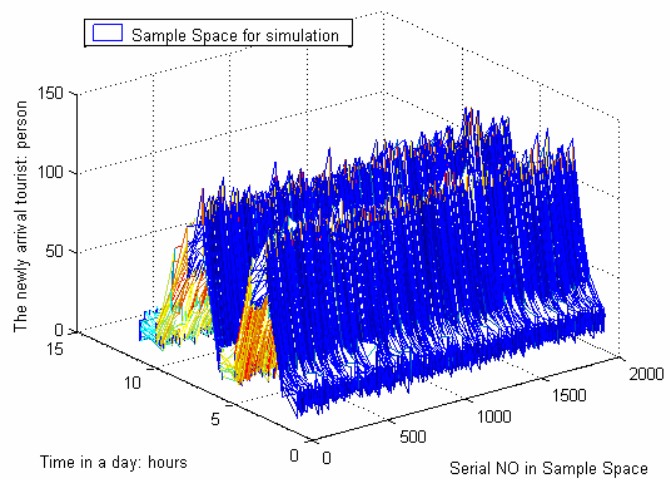


Fig 10: the Sample Space generated for simulation according to assumptions

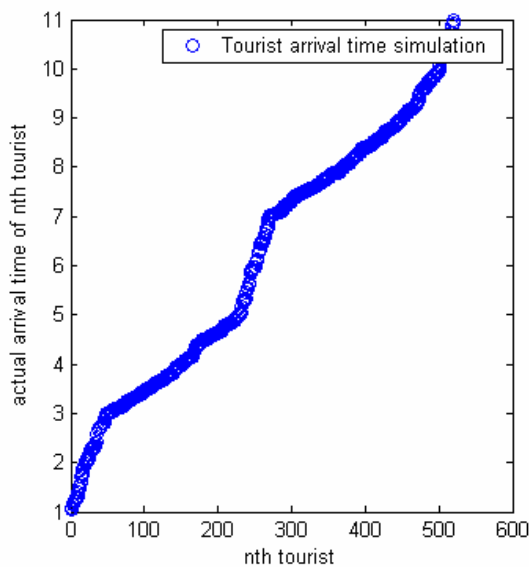


Fig. 11 Poisson tourist arrival time

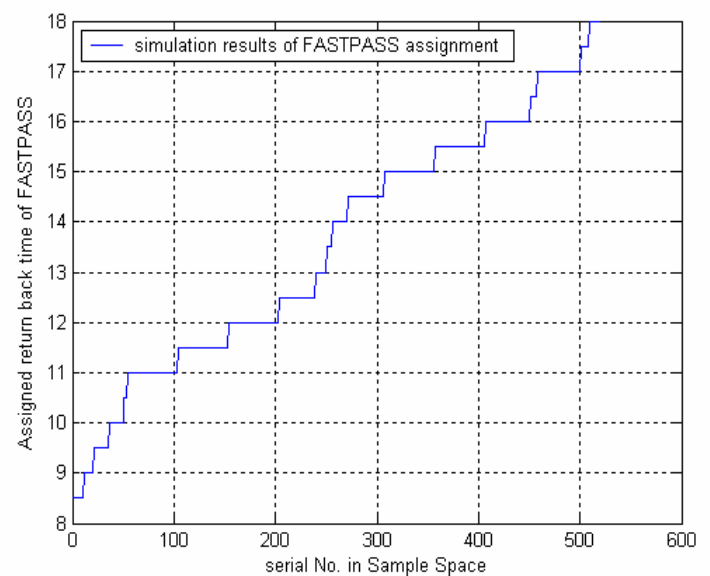
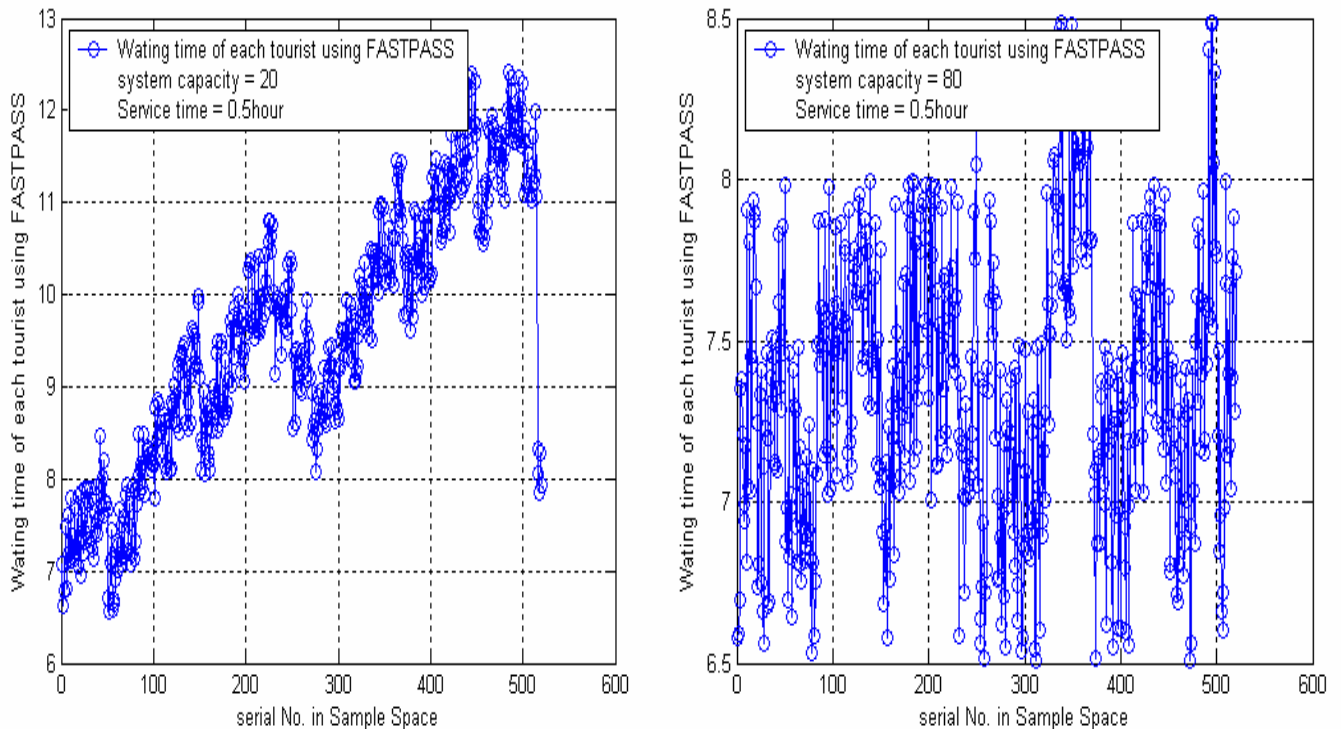


Fig. 12 The shifts arrangement for tourists in fig. 11 by ‘Greedy Algorithm’

The following fig.13 and fig.14 shows delay time of tourists, which was greatly affected by algorithm use



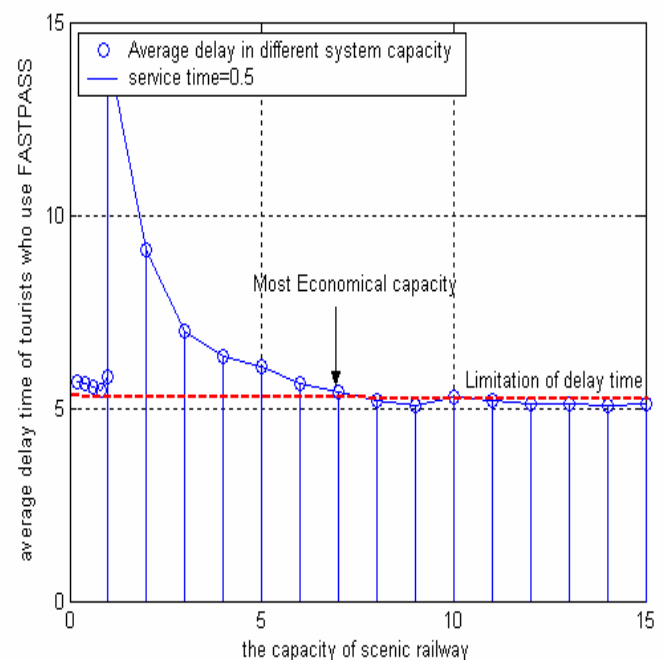
railway could carry once. From the figure we could notice that system capacity would phenomenally affected distribution of delay time. So we think about to find their relationships, and even to optimize. (service time is cycle interval)

4.3 Optimization of system capacity

We still use the same cost function, but simplify it to assume $c_2 t_2$ is negligible, so the following

simulation we simply use $E\{c_1 t_1\}$ as cost function,

and use average delay time (ADT) as our main parameter. From fig.15 we discover that the gains brought by larger system capacity would have an upper limit. So larger system capacity do not always mean smaller ADT, in view that the seat of scenic railways generally do not exceed 100 seats and larger railways would be expensive we adopt system capacity = 70 as an economical value in our model.



4.4 Optimization of cycle interval

Fig. 15 System capacity vs. average delay time

Cycle interval, i.e. the shift interval of scenic railway, could also be a parameter that seeking optimization. Fig. 8 shows the result of our simulation, which show that unfortunately ADT(cycle interval) is monotonous without extremes in an interval. However, we seek to trade off between ADT and economic gain of amusement park, in view that increase cycle interval cause incensement of ADT, while decrease economic gain. Fig. 17 demonstrates our solution to this trade off which we obtained from simulation, with the assumption that each shift of scenic railway cost 100\$, a single ticket is sold at price of 10\$. The trade off point is the intersection with cycle interval = 4.67 min.

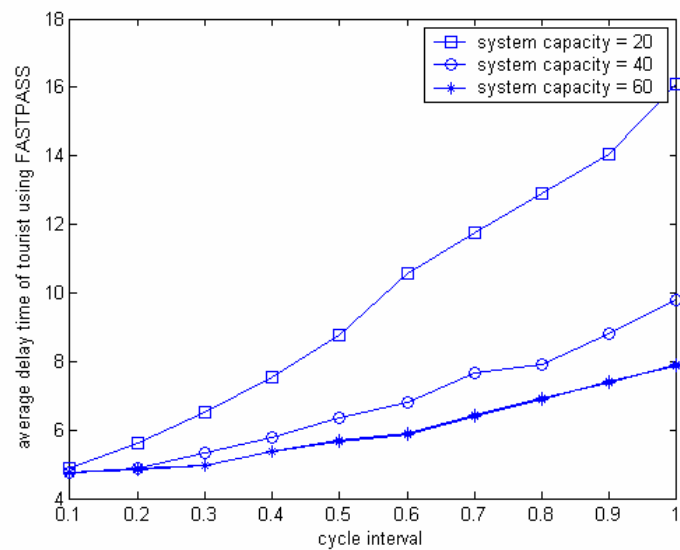


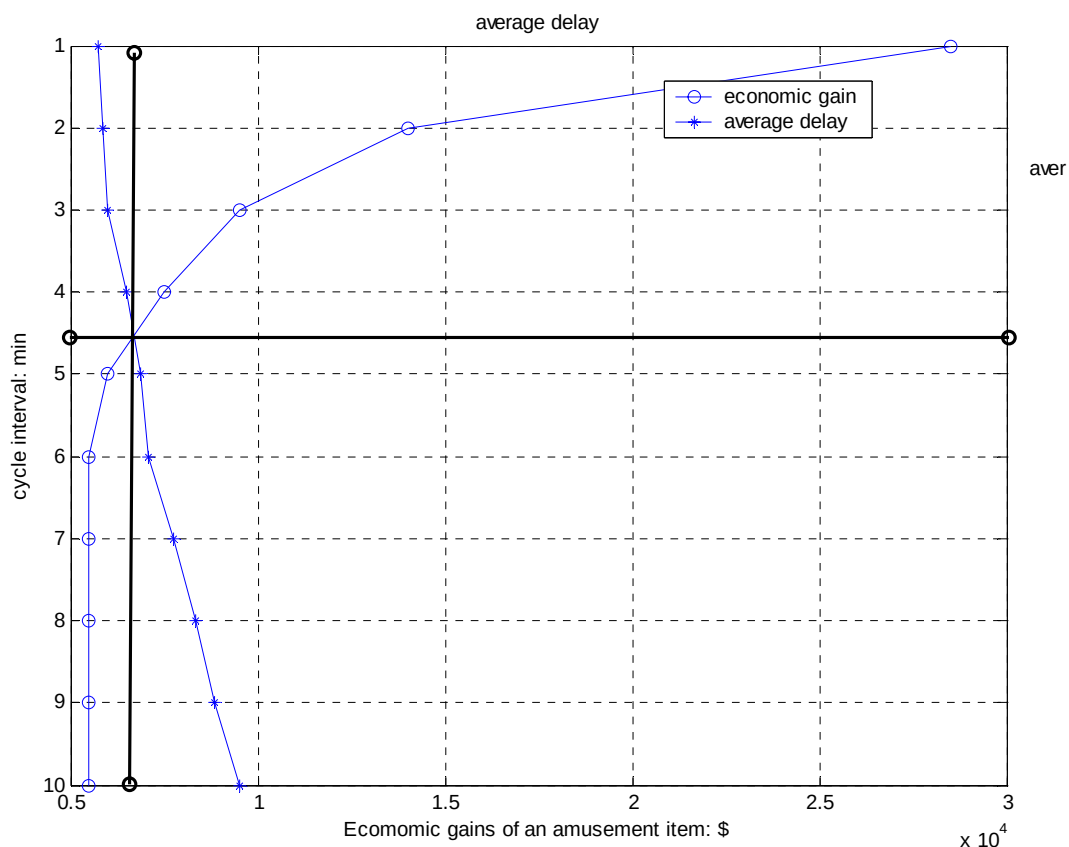
Fig. 16 Cycle interval vs. ADT

4.5 Comparison with Regular Lines

If, and how much FASTPASS wins over ordinary line, is also our focus. Using result got from previous part of this paper, also simulation data, we could calculate that

On average ADT of FASTPASS is 11.2% shorter than ordinary lines, and in terms of the cost function, FASTPASS is 21.3% less than the later. This test demonstrates that our FASTPASS algorithm indeed counts.

Fig. 9 Trade off between economic gains of amusement park and average delay time



5 Strengths and Weakness

Strengths:

1. For the tourists' arrival having certain rule, our model can relatively accurately assign the tourists to reduce the waiting time.
2. The profitability of the tourists being in standby line is also considered, which is more popular by people, because the standby line exists in spite of how healthy Quick Pass system is. As the part of all the tourists, the tourists waiting in standby line is to some extent affecting the revenue of the park.
3. We keep the fixed elastic amount of the tourists, here, is $0.2M$, to make sure that the number of the tourists being in each operating cycle of the item gets to a fixed amount; this can increase the revenue of each operating cycle of the item.
4. Our model uses the statistic data to predict the transformation of the queue length during a day's work time. It suggests the tourists at a rush time come back at a low time. So "the traffic flow" may become smoother.

Weakness:

1. The actual length of the queue may be much different from our prediction. So the tourists come back later may not wait shorter in line.
2. Different people have different c_1 and c_2 . But we simply assume that they have the same c_1 and c_2 . So a curious tourist with his c_1 and c_2 much different from others' can not benefit from the quickpass's suggestion.

6 Further works – A better QuickPass

1. All the models considered above assumption the tourist is passive, they only return during the time told by the "QuickPass" slip, but in the really world, the tourist has the right to choose the 2 lines to entrance. If the tourist holding the slip knows the length of the standby line is so short that s/he being in the line will entrance the attraction earlier, thus, it is possible that s/he will be in the standby line. So, a healthy "QuickPass" system should tell the tourist the current length of the standby line and the approximate waiting time if the tourist joins in the standby queue at once. The tourist can compare the 2 kind of time told to choose which line. Based on it, we can create a better relatively model.
2. Because each tourist has different time scheme, the tourist may be not content to the assignment from the system. For example, a tourist only has 1 hour's time, but s/he is told to return after 2 hours at least, thus s/he only waits in the standby line to paly. If the system considers the tourist's limited time, then the system can assign the tourist to return during the limited time. A better model may be based on it.
3. The park can take all the items in the park as a whole, and generally, the tourist goes to a park for playing several favorable items, people often hope to play all-wanting items during much shorter time. For realizing the purpose, we can design the system that the slip not only tells the returning time

interval, but also tells the tourist the information about other items, for example, suggests the tourist to can paly some certain items after waiting much shorten time, that can make the tourist use reasonably t_1 (in our model) to the max whole utility of the tourist. Of course, for the simplification of the operation, we can create a database to store the information of each tourist, then when the tourist visits the park, and uses the system, the slip will display additional information about the items the tourist often play.

4. For increasing the ability of dealing with the exigent state, for example, on weekends or at holidays, suddenly coming a large of the tourists, we must design a system called **Emergency-dealing System** in the “QuickPass” system; the system can only service the sudden large amount of people. We can set a switch to evaluate when to start the system, for example, when the “QuickPass” system counts the amount of the people inserting ticket per time is more than a fixed value defined by statistical methods.
5. In our model, we assume the system makes the same amount of the tourists to return during the same interval. Here, there is a problem, that is, Can we assign the different amount tourists to return each interval? If can, our objective is how to assign the tourists, how many and what persons are assigned to return during the first interval, what about the second interval? The third interval? to make minimize the function $\phi = E \sum_k (c_{k_1} t_{k_1} + c_{k_2} t_{k_2})$. If we can find the rule assigning the tourists, we can solute the complain that the tourists are told to return after 4 hours, while the tourists later are told to return the same ride only after 1 hour or so.

7 A non-technical summary for executives:

Whether the “QuickPass” system is popular to the tourists is determined by the degree that the tourists feel the waiting time reducing. So, one of the criteria evaluating the alternative schemes is to make the tourists really to feel the waiting time reducing, that is, the difference between the waiting time in the usual line and that in the “QuickPass” line is relatively evident.

Another factor must be considered, that is the profitability of the item allowed to use the “QuickPass” system. Because some times the scheme only considers the waiting time of the people, so that increasing the operating cost, but the revenue is not steadily rising, even it is often less than cost.

The ability dealing with the emergency, for example, if suddenly a large amount of people come during a very short interval, if the system does not reasonably deal with the problem, the case that the waiting time in “QuickPass” line is as long as the regular line or even longer will occur.

8. References

- [1] Qiyuan Jiang, Mathematic Model, P457, Advanced Education Public, 1993, CHINA
- [2] Theodore S. Rappaport, Wireless Communication: Principles and Practice, P605-610, Prentice Hall, 2002, USA
- [3] Ying'ai Gan, Operation Research, P342, the Public Tsinghua University, 1990, CHINA
- [4] Jianying Hu, Tai Feng, Practical Statistics, 1996, Perking University Press.
- [5] Qian Songdi, the research of operation, Tsinghua Publish Press;
- [6] Yiyuan Li: Western Economics, Advanced Education Public, 2001, CHINA
- [7] Kishor S. Trivedi, Probability & Statistics with Queuing and Computer Science Application, Prentice – Hall, 1982.
- [8] Xiru Chen, Probability Theory and Mathematical Statistics, Press of University of Science and Technology of China, 2002
- [9] Zhaoben Fang, Qiupo Miu, Stochastic Processes, Press of University of Science and Technology of China, 1992
- [10] Qingyuan Computer Studio, Matlab 6.0: Fundamental and Application, China Machine Press, 2001.
- [11] Goodman R. Iversen, Mary Gergen, Statistics: the Conceptual Approach, Springer, CHEP, 2000.

- [1] <http://probbstat.nuk.edu.tw/web/pdf14.htm>
- [2] <http://allearsnet.com/tp/fastpass.htm>
- [3] <http://www.mouseplanet.com/al/docs/fast.htm>
- [4] <http://www.dlp-guidebook.de/HintsInfos/Fastpass.htm>.
- [5] <http://www.dlp-guidebook.de/HintsInfos/Fastpass.htm>
- [6] <http://www.disneyland.com>